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An Integrated Approach Using Ensemble Machine Learning and Deep Learning for Loan Approval Prediction

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***Abstract*—The suggested loan prediction system overcomes the drawbacks of conventional selection processes by utilizing machine learning (ML) to enhance the identification of creditworthy borrowers. A Random Forest (RF) model to predict loan is evaluated in our study along with data pretreatment using the Synthetic Minority Over-Sampling Technique (SMOTE) to solve the class imbalance. Nine ML models in all, including Deep Learning (DL) approaches, ensemble learning strategies, and conventional classifiers, were used in this work to reliably and successfully forecast loan acceptance results. According to a number of characteristics, the dataset utilized in this study aims to predict whether a loan application will either be approved or denied. The RF model obtained 94.10% of F1-Score, an Accuracy of 94.12%, Precision and Recall values of 94.24% and 94.12%, respectively, following the use of SMOTE for maintaining equilibrium of data.These findings demonstrate RF's effectiveness in managing unbalanced datasets and its capacity to forecast loan approval outcomes. Because they automate the approval process, lower human error, and increase decision-making efficiency, loan prediction models are essential to financial organizations.**

Keywords—Loan prediction, Machine learning, Deep learning, SMOTE, Random Forest.

I. Introduction

The banking business is critical to ensuring a country's financial stability; hence, in the majority of nations, it is governed by comprehensive regulatory systems. Lending is one of the primary activities of banks, and the interest revenue generated from loans accounts for a considerable measurement of their assets[1].Nonetheless, the existing loan acceptance procedure has significant obstacles, particularly in terms of efficiency and accuracy, given its reliance on manual methods. These processes delegate responsibility for analyzing applicants' eligibility and loan default risk to individual bank managers. These manual processes have far-reaching consequences, ranging from huge financial losses for banks to systemic disruptions that could hurt the economy as a whole. Throughout banking history, the loan acceptance procedure has focused on the difficulty of differentiating creditworthy debtors from a sizable candidate pool [2]. Predicting loan default effectively has become increasingly important in contemporary financial systems. Failure to foresee wide-ranging effects of loan defaults include banking crises [3]. To identify loan defaulters manually has proven to be a difficult undertaking, considering the complexity of today's financial industry and the growing request for loans. ML methods, which allow computers to proactively identify features and generate data-driven estimates, have arisen as a prospective way to figure out the probability of loan insolvency [4]. These algorithms dig through users' transaction records and social profiles, revealing behavioral patterns and general characteristics which give insights about prospective repayment tendencies [5].

Furthermore, DL methodologies, including deep neural networks, have gained significance in this domain due to their proficiency in gathering intricate, non-linear correlations within the data [6]. Further improving the assessment of the probability of defaulting on loans for
specific consumers. As a result, grounded in ML and DL techniques have been intensively studied to give a thorough assessment of loan default risk. In this environment, banks' importance in assessing credit risk cannot be overstated. Assessing borrowers' credit histories before giving loans is critical for sorting among potential defaulters and non-defaulters. Creating a precise estimating system by hand is a difficult task because of the variety of this sort of thing and the growing demand for loans. Although several previous studies have explored ML models for loan failure estimation, relying solely on one classifier is not enough for practical use.

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To improve loan acceptance processes, we offer an ensemble technique that makes use of top-performing ML algorithms. Loan acceptance is a key procedure for financial organizations, requiring a thorough assessment of numerous aspects such as income, social standing, and citizenship to assess loan qualification. Timely loan payback is critical to the performance of banks, which need an accurate evaluation of borrowers' repayment potential [7].

ML has been studied in a number of studies to enhance loan acceptance prediction. Standard decision trees were beaten by RF classifiers, which also developed an ensemble model that raised accuracy in predicting from 80% to 94% and suggested a stacking-based method that significantly enhanced joint loan acceptance.[8]. Used cutting-edge ML methods, specifically RF, to predict loans that fail in developing nations. Multiple models have been provided in this field by previous studies, but there hasn't been a comprehensive analysis that incorporates both ML and DL techniques. Moreover, the majority of existing systems have not proven to be accurate enough for practical applications. [9]. Ensemble model that forecasts using a combination of ML and DL for loan defaults accurately and efficiently, improving modern banking's dependability.

The following are the research's major contributions:

- **Data Balancing with SMOTE:** By addressing class imbalance in loan prediction, this work uses SMOTE to enhance model performance and guarantee more equitable forecasts for all classes.
- **Comprehensive Model Evaluation:** Nine ML models, combining ensemble methods and DL techniques, are evaluated using Accuracy, Precision, Recall, and F1-Score, providing a thorough assessment of model performance.
- **Visualization of Model Performance:** RF was the best-performing model, with 94.12% accuracy, 94.24% precision, 94.12% recall, and 94.10% F1-score. To further understand its predictive capacity and do additional performance analysis, visualizations of the ROC curve and confusion matrix are employed.
- **Model Interpretability:** Leaf node analysis is used to show the RF model's decision-making process, improving interpretability and transparency for practical loan prediction applications.

These additions demonstrate the study's comprehensiveness in terms of technical assessment and model interpretability, which makes the research useful for enhancing loan prediction systems.

There are **three** components to this study. The **Literature review** section provides a summary of previous research in the field of interest. The methodology employed in this study is detailed in the **Materials and Methods** section, which also includes an analysis of the experimental data and a comprehensive explanation of the results. The final part, **Conclusion**, summarizes the key findings and offers recommendations for more research.

II. Literature Review

Modern RF algorithms were employed by Bhargav and Sashirekha (2023) for contrasting ML approaches for granting loans estimation. Loan estimation samples from the Kaggle library were used to calculate reliability and revenue. The RF technique fared better than the normal tree of decisions in an average of 20 instances, with 79.44% consistency and 21.03% loss as opposed to 67.28% consistency and 32.71% attrition for the former. A p-value of 0.33, at a 95% confidence level, was obtained from statistical analysis using a separate sample T-test, indicating no appreciable variations between the approaches. This study found that RF outperformed decision chains as a potential indicator of loan clearance [7].

In order to approve financial institution risks, Wang et al. (2023) developed a stacking-based model. The optimal model is chosen by comparing performance. Additionally, they used DL to create a bank clearance model using unbalanced data, extracting features with CNN, and balancing sampling with counterfactual augmentation. Experiments on real data show that optimizing the auto collaborative acceptance of loans raised by about 6% using a finance prediction model based on bank model parameters [8]. Dasari et al. (2023) created an ensemble model that incorporates various ML methods with voting classifiers and bagging techniques. Predicting loan eligibility was the primary goal. This model outperforms current techniques by improving accuracy and requiring less human labor and processing time. Comparing the experimental findings to the prior model, there is a performance increase from 80% to 94% [10].

Abdullah et al. (2023) employed a variety of ML techniques to forecast bad debts in lenders in developing nations. In an investigation of data from 322 banks across 15 countries, modern ML models—in specific, RF—performed better than linear approaches with a success rate of 76.10%. In the ineffective loan projection, bank multiplicity was found to be a major predictor, surpassing macroeconomic variables [11]. In order to use conventional methods to better accurately assess bank loan threats, the effectiveness of ML algorithms was examined. When it came to detecting bank loan threats, Alsaleem and Hasoon (2020) discovered that Multilayer Perceptron (MLP) functioned better than RF, BayesNet, Naive Bayes (NB), and DTJ48 algorithms. Familiar measures were used to assess the efficacy of the model on a dataset of 1,000 loans and their repayment conditions [12].

The performance of two algorithms, RF and DT, was examined by Ghatasheh (2014). They found that the RF algorithm worked better than any other algorithm for forecasting credit risk during their analysis [13]. In their comparison of popular ML models, Shoumo et al. (2019) discovered that SVM performed better than LR and RF. Their results verified that the LR model can determine which target customers are most likely to approve a loan. According to the approach, banks shouldn't restrict credit availability to wealthy people [14]. Both the training and testing data sets were derived from the bank client information. The test sample lacks the desired feature but has roughly 300 rows and 12 columns, while the training dataset has roughly 600 rows and 13 columns (Jency et al. 2018) [15].

Anand et al. (2022) utilizing ML models for guarantee finance, he estimated loan actions via a dataset of 850 paperwork. Their model included DT, RF, ET, CatBoost (CB), Light Gradient Boosting (LGB), and Extreme Gradient Boosting (EGB). Their results showed that the ET and RF were more accurate at establishing loan acceptance [16]. A

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Kaggle source provided the loan data used by Blessie and Rekha (2019) (Chatterjee, 2021). They employed four models—LR, DT, SVM, and NB classifiers—to predict loans. Their results showed that NB was more accurate in forecasting credit availability [17][18]. In order to determine loan eligibility, Kumar et al. (2022) [19] gathered a dataset of 614 items from a freely accessible source. A range of ML techniques, such as RF, DT, KNN, SVM, and DT with AdaBoost, were employed. In a similar vein, (Dosalwar et al., 2021) [20] gathered a Kaggle loan forecast dataset. It was discovered that the LR model predicted loan eligibility more accurately.

III. MATERIALS & METHOD

"Fig. 1" shows the technique, which starts with dataset preprocessing, which includes feature scaling, exploratory data analysis (EDA), outlier removal, managing missing and null values, and class balance using SMOTE. Training accounts for 70% of the data, while testing accounts for 30%. To capture complicated patterns, deep learning architectures (LSTM, MLP, PLSTM) are integrated with machine learning models (RF, LGBM, SVM) and ensemble approaches (soft and hard voting). Model performance is optimized via hyper parameter tuning, and prediction accuracy and dependability measurements are used to choose the top-performing model.

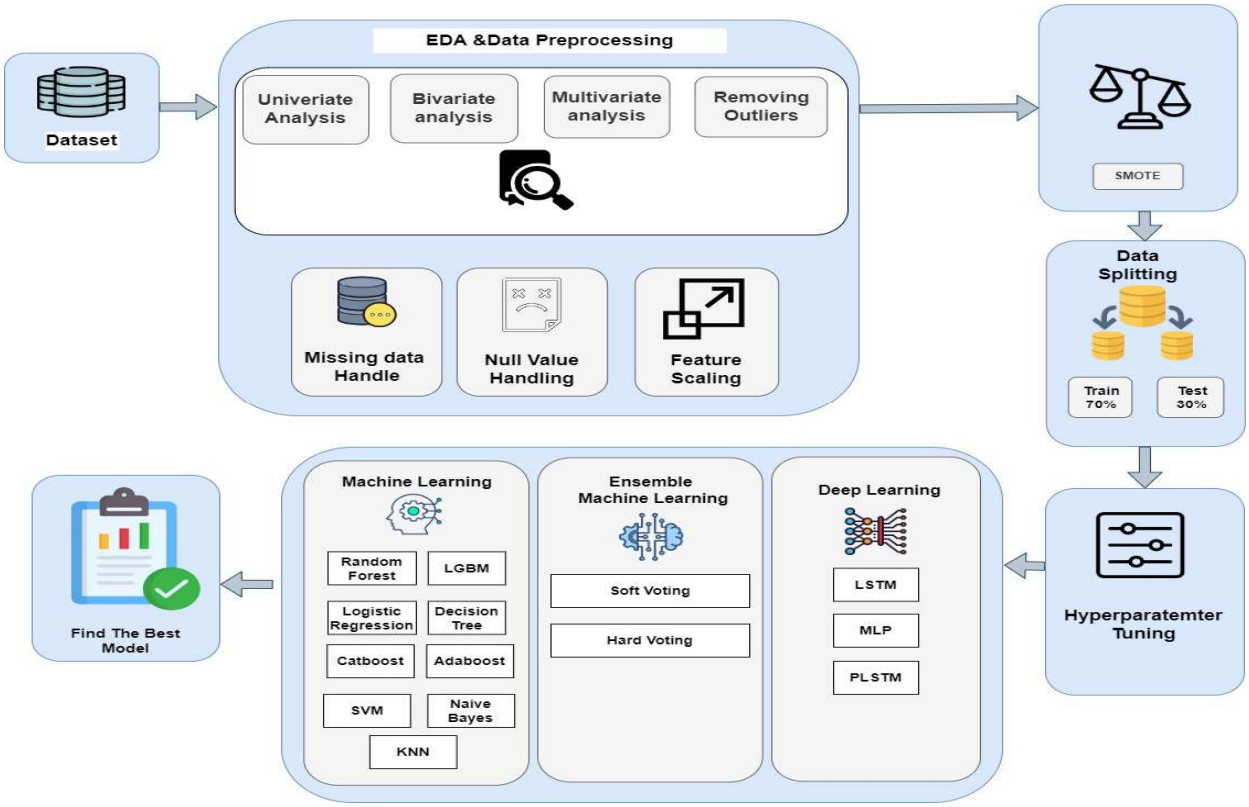


Fig. 1: The operational procedure of the proposed technique

A. Dataset

Table I provides an overview of the loan dataset data, which was obtained from Kaggle. As explained in the same table, the table includes a number of aspects. visualization attribute, where the target attributes of the loan dataset are represented by the YES and NO statuses. A business wants to use the details that clients provide when filling out an online application form to automate the loan qualifying process (in real-time). It has released this dataset, which consists of 13 columns and 614 rows, in order to accomplish this.

TABLE I. **PRESENTING THE DATASET**

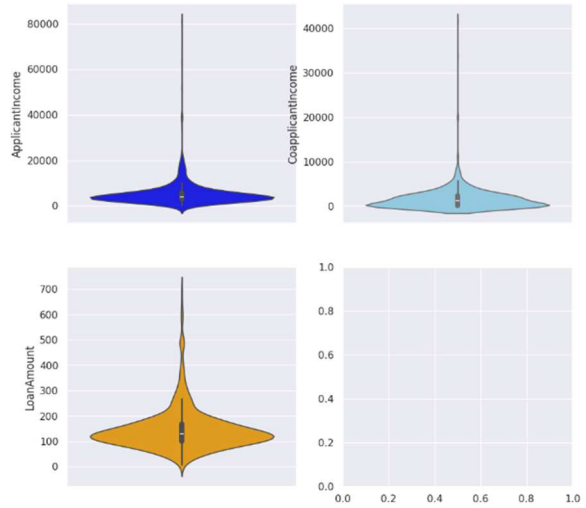
Feature Name	Feature Details	Type of Data
Loan ID	Loan reference number	Integer
Sexual orientation	Applicant gender	Character
Wedded	Applicant marital status	Character
Dependents	Number of family members	Integer
Education	Graduate/ Not Graduate	String
Self-Employed	Applicant employment status	Character
Income of applicant	Applicant's monthly salary/income	Integer
Income of Co-applicant	Additional applicant's monthly	Integer
Amount of Loan	Loan amounts in thousands	Integer
Term of Loan Amount	The loan's repayment period (in days)	Integer
Credit History	Records of previous credit history	Integer
Property Area	The location of property	String
Loan Status	Status of loan	String

B. Data Preprocessing

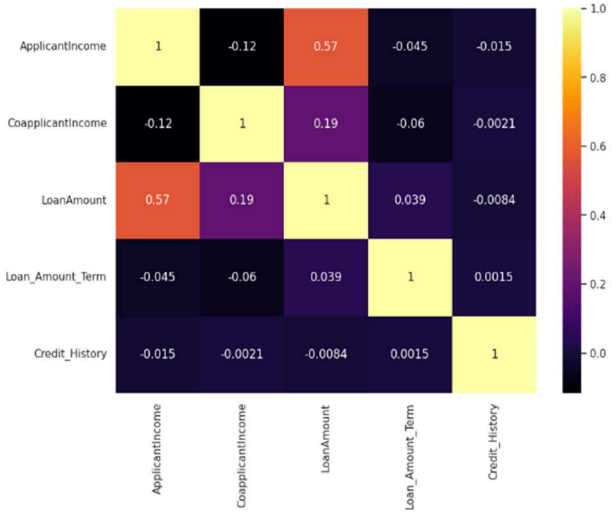
The dataset was cleaned up for machine learning by eliminating outliers, leveling skewed distributions, encoding categorical features, and resolving missing values for

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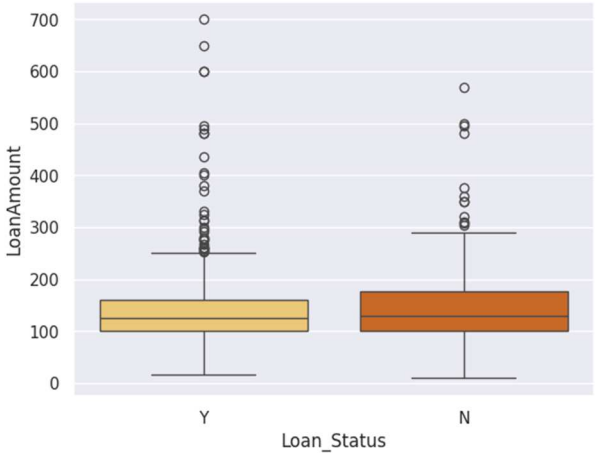
constant as well as categorical variables, respectively, the mean and mode were used to impute missing values. In order to normalize skewed data, outliers were removed using the IQR technique and one-hot encoding was used for categorical characteristics. A balanced dataset was produced by using SMOTE to correct the target variable's class imbalance. Min-Max normalization was used to scale the data, and it was divided into training (70%) and testing (30%) groups. The preparation procedures and how they affect the dataset are shown in Fig. 2.



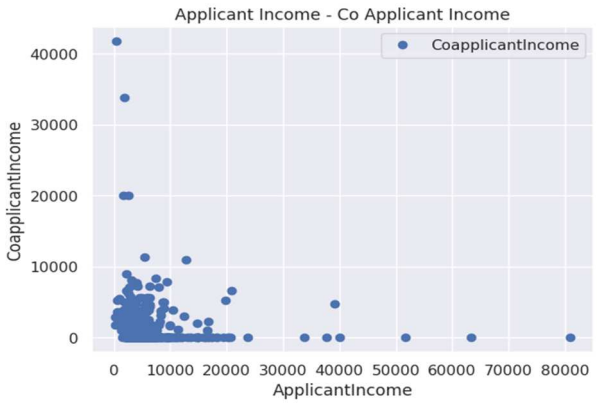
There are outliers and a negative skew in the Loan Amount Term distribution.



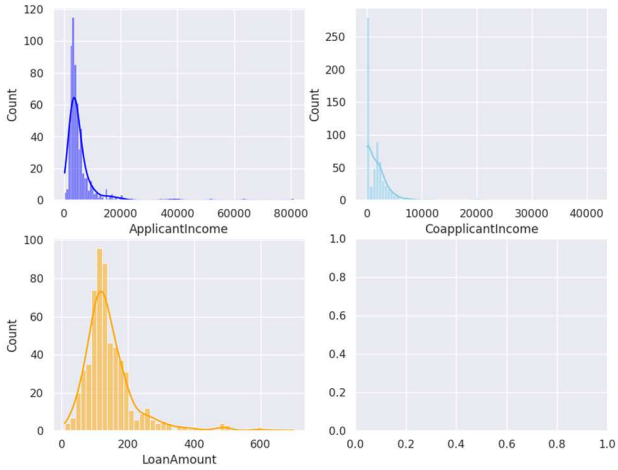
The applicant's earnings and the loan amount are positively correlated.



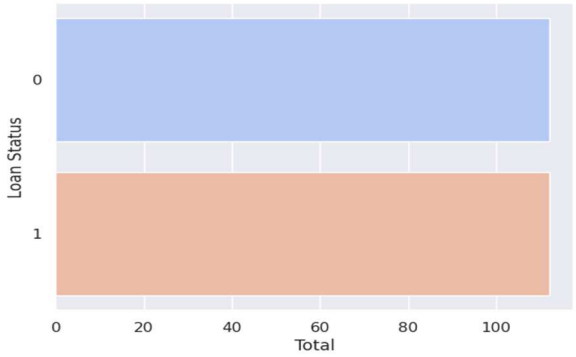
There are many outliers in the co-applicant distribution of earnings, which is likewise favorably skewed.



Negatively correlated earnings of the candidate and the fellow applicant



When log transformation is applied, the distribution is far better than it was before.



There is currently a balanced distribution of loan status.

Fig. 2: Some visualization of dataset in preprocessing

IV. RESULT ANALYSIS

Table II summarizes the assessment of numerous deep learning, ensemble, and ML models' performance, showing notable variations in F1-score, accuracy, accuracy as well as memory. RF beat all other machine learning models with an accuracy of 94.12%, sharpness of 94.24%, recall of 94.12%, and F1-score of 94.10%, indicating its resilience. As the number of nodes increases, RF accuracy rises sharply, culminating at 13–14 nodes with an accuracy close to 94.5%. This is seen in Fig. 3, which shows RF accuracy versus the maximum left nodes.

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TABLE II. TABLE TYPE STYLES

Machine Learning Models				
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Random Forest	94.12	94.24	94.12	94.10
KNN	86.76	86.99	86.76	86.70
SVM	77.94	80.34	77.94	77.73
Logistic Regression	79.41	80.56	79.41	79.36
Naïve Bayes	67.65	67.67	67.65	67.42
Decision Tree	76.47	76.64	76.47	76.49
AdaBoost	72.06	72.78	72.06	72.03
LightGBM	83.82	83.82	83.82	83.81
CatBoost	92.65	92.95	92.65	92.61
Ensemble Machine Learning				
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Soft Voting Classifier	88.24	89.37	88.24	88.08
Hard Voting Classifier	91.18	91.70	91.18	91.11
Deep Learning				
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
LSTM	83.82	83.87	81.24	82.54
PLSTM	76.47	72.22	81.25	76.47
MLP	76.47	75.00	75.00	75.00

Accuracy plateaus or slightly declines after this, suggesting that the benefits of increasing tree complexity fade. “Fig. 4” illustrates the RF classifier's Confusion Matrix, which shows how well it can categorize instances. The model performed well in both categories, properly classifying 35 true negatives (class 0) and 29 true positives (class 1) out of 68 total samples. Nevertheless, it incorrectly identified three cases as false negatives and one as a false positive.

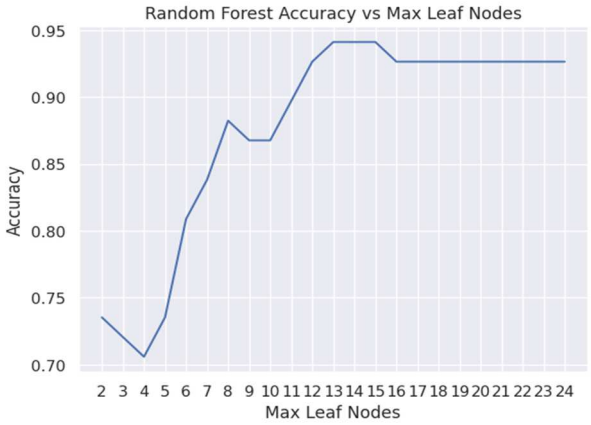


Fig. 3: Random Forest Accuracy vs Max Left Nodes

The RF model's excellent accuracy, recall, and F1-score values further support the model's good sensitivity-specificity balance, as seen by these few misclassifications. This illustrates how well the model manages class disparities and produces accurate forecasts. With an accuracy of 92.65% and similar precision and recall, CatBoost was a strong contender. KNN came in second with a mediocre performance (accuracy: 86.76%, F1-score: 86.70%). Naïve Bayes showed the least efficacy with an accuracy of 67.65%, while Logistic Regression (accuracy: 79.41%) slightly exceeded SVM (accuracy: 77.94%).

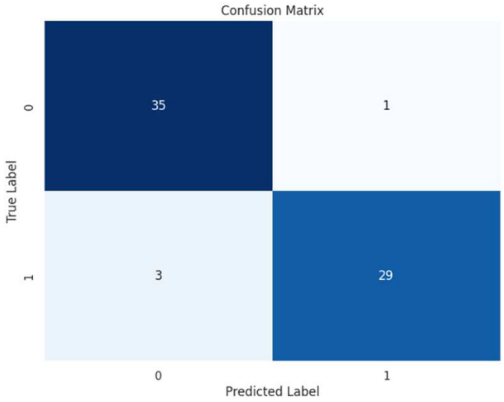


Fig. 4: Confusion Matrix of Random Forest

Through model aggregation, ensemble models such as the Hard Voting Classifier (accuracy: 91.18%, F1-score: 91.11%) outperformed the Soft Voting Classifier (accuracy: 88.24%). The ROC Curve of nine classifiers is shown in Fig. 5, where RF has the maximum AUC, confirming its superior discriminative abilities. CatBoost and the Hard Voting Classifier follow.

With an accuracy of 83.82% and an F1-score of 82.54%, LSTM outperformed other deep learning models in terms of capturing sequential patterns. With accuracies of 76.47%, PLSTM and MLP, on the other hand, showed limited applicability and needed architectural changes. These findings highlight the efficiency of ensemble and tree-based approaches in reaching high prediction accuracy, but deep learning models and Naïve Bayes need to be improved for improved performance.

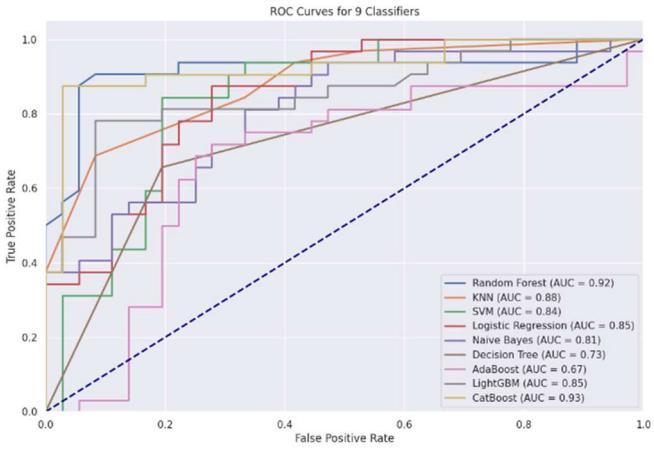


Fig. 5: ROC Curve of 9 ML Classifier

Table III highlights how well the suggested RF model performs when compared to other studies conducted in related fields. With an accuracy of 94.12%, the suggested model performs noticeably better than RF, which Bhargav & Sashirekha (2023) used to obtain an accuracy of 79.4% on the Kaggle Loan Prediction dataset. The resilience of the suggested model was further shown by Abdullah et al. (2023), who used Extremely Randomized Trees (ET) on the same dataset but only obtained 76.1% accuracy. Using a combination of CatBoost, LightGBM, and Extreme Gradient

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Boosting (EGB) on the Home Credit Default Risk dataset, Anand et al. (2022) achieved an accuracy of 86.2%, which is impressive but falls short of the outcomes of the suggested model. Kumar et al. (2019) and Blessie & Rekha.

TABLE III. **COMPARING THE OUTCOMES OF VARIOUS CURRENT WORKS**

Author	Dataset	Mathod	Accuracy
Bhargav & Sashirekha,2023 [7]	Kaggle Loan Prediction	RF	79.4%
Abdullah et al.,2023[11]	Kaggle Loan Prediction	ET	76.1%
Anand et al.,2022[16]	Home Credit Default Risk(Kaggle)	CB, LGB, EGB	86.2%
Blessie & Rekha,2019[17]	UCI Machine Learning Repository	MP	80.4%
Kumar et al.,2022[19]	Custom Dataset (collected from financial institutions)	DT, KNN, SVM	72%
Proposed Model	Kaggle	**RF** (ML)	94.12%

Using Multilayer Perceptron (MP) and a combination of Decision Tree, KNN, and SVM, respectively, on various datasets, (2022) found accuracies of 80.4% and 72%; nevertheless, neither method can predict as well as the suggested model. This significant increase makes the Random Forest algorithm a state-of-the-art solution for the dataset in question and demonstrates its effectiveness and appropriate tuning in the suggested work.

V. CONCLUSIONS

Using important measures like Accuracy, Precision, Recall, and F1-Score, we assessed a number of ML models in this article, including RF, Hard Voting Classifier, and LSTM. With an accuracy of 94.12%, the Random Forest model beat the other classifiers, proving its usefulness in this challenge. Even though the Hard Voting Classifier, which combines several classifiers, had somewhat worse results, it was still useful since it gave an alternative viewpoint on ensemble learning. Despite demonstrating promise in deep learning applications, the LSTM model performed comparatively worse than more conventional models. Future research will concentrate on improving model interpretability using Explainable AI (XAI) methods like SHAP and LIME, which can boost prediction confidence and offer deeper insights into model decisions. To improve performance, it will also be crucial to investigate hybrid strategies and adjust hyperparameters, especially for deep learning models. Since model transparency and dependability are crucial in decision-making processes, real-world applications will substantially benefit from these advancements, particularly in industries like healthcare, finance, and autonomous systems.

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